

Shihab Muhtasim

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GitHub | LinkedIn | Portfolio

ABOUT ME

Erasmus Mundus Joint Master's student in computer vision and image processing, currently based in Madrid. Focused on video analytics, object detection and classification, and multi-camera visual understanding. Experienced in building and evaluating deep learning systems for detection, segmentation, re-identification, and 3D scene analysis through academic projects and research.

INTERESTED FIELDS

Computer Vision, Video Analytics, Object Detection and Recognition, 3D Reconstruction, Image Processing, Re-Identification, Explainable AI, Natural Language Processing, Machine Learning.

EDUCATION

Erasmus Mundus Joint Master's Programme (IPCVAI) Budapest, Madrid and Bordeaux
Master of Science in Artificial Intelligence for Image Processing and Computer Vision Sep. 2025 – Current
Mobility structure:

- Sep 2025 – Jan 2026 – Pázmány Péter Catholic University, Budapest, Hungary
- Feb 2026 – May 2026 – Universidad Autónoma de Madrid (UAM), Madrid, Spain
- Sep 2026 – Jan 2027 – University of Bordeaux (UBx), Bordeaux, France
- 6 months – Internship in academia or industry

Website: IPCVAI **ECTS:** 120

BRAC University Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering Jun. 2021 – Nov. 2024
CGPA: 4.00/4.00 **Credits:** 136
Website: BRAC University

WORK EXPERIENCE

Adjunct Lecturer Dhaka, Bangladesh
Department of Computer Science and Engineering, BRAC University Feb. 2025 – Jun. 2025
Roles: Delivered undergraduate-level lectures in courses such as Data Structures, OOP, OS, Microprocessors, and AI. Designed and evaluated assignments, quizzes, and examinations, supervised student projects, and provided academic mentoring.
Link: University Profile

Teaching Assistant Dhaka, Bangladesh
Department of Computer Science and Engineering, BRAC University Oct. 2023 – Aug. 2024
Roles: Provided weekly consultations, assisted professors with lab sessions, graded assignments, conducted oral exams, and administered final lab exams for the Data Structures course.
Link: Certification

Head of Web Development Dhaka, Bangladesh
Institute of Young Talents (IYT) Sep. 2023 – Oct. 2024
Roles: Led the web development team, guided several projects, recruited new members, and developed websites.
Link: Reference

HONOURS AND AWARDS

- **Erasmus Mundus Scholar**, European Commission (2025): Selected for multiple Erasmus Mundus Joint Master's programs with fully funded scholarships from the European Commission. Programs awarded: IPCVAI, CoDas, EMJM Imaging.
- **Chancellor's Gold Medalist**, BRAC University (Feb. 2025): Chosen as the chancellor's gold medalist out of 4600 graduates at the 16th convocation as the top graduate. **Link:** Certification
- **Valedictorian of the 16th Convocation**, BRAC University (2025): Selected to represent the graduating class and deliver the valedictorian speech. **Link:** Video

RESEARCH PROJECTS

Evaluation and Improvement on 3D Reconstruction Based on NeRF models Jan. 2026 – Current
This project involves the study and evaluation of existing Neural Radiance Field (NeRF)-based methods through implementation on suitable datasets and analysis of performance to support benchmarking and reproducible research with a goal of extending the project to include some improvements of models where weakness is found.

Analyzing and Predicting Trends in Contemporary Social Discourse through Hashtag 2023 – 2024
Analyzed social media hashtag popularity by modeling information diffusion and user interaction networks, and predicted trend popularity using time-series forecasting with LSTM, network science, graph theory, and NLP techniques.

Link: IEEE Paper

Polyp Segmentation and Identification from Endoscopic Images Using Deep Learning 2025
Supervised research that proposed a deep learning-based approach for automatic colorectal polyp segmentation in endoscopic images, evaluating convolutional and transformer-based models to improve detection accuracy.

Link: IEEE Paper

RECENT PROJECTS

Cross-Camera Re-Identification on Urban Elements Feb. 2026 – May 2026
Kaggle Urban Elements ReID Challenge 2026 (UAM). Cross-camera retrieval system for urban objects across non-overlapping surveillance cameras using a Part-Aware Transformer with ensemble methods and camera-adversarial training; achieved **0.15884 mAP@100** (54.7% relative improvement over baseline). We are at the top 10 on the leaderboard as well. **Link:** GitHub

CLIP Spatial Reasoning Explainability and Failure Analysis Feb. 2026 – May 2026
Investigated spatial reasoning failures in CLIP ViT-B/32 through seven structured experiments on the Visual Spatial Reasoning dataset using Grad-CAM attribution, layer-wise text embedding analysis, and comparative evaluation against BLIP. **Link:** GitHub

MultiView 3D Reconstruction Feb. 2026 – May 2026
End-to-end MATLAB pipeline for Euclidean 3D scene reconstruction from uncalibrated images — implementing Zhang's camera calibration, SIFT feature matching, RANSAC-based homography and fundamental matrix estimation, multi-view DLT resection, projective bundle adjustment, and Essential matrix decomposition with chirality-based solution selection. **Link:** GitHub

TECHNICAL SKILLS

Programming Languages & Web Development: Python, MATLAB, Java, C++, C, Laravel, React, HTML, JavaScript, CSS, SQL, MySQL, PHP

Libraries & Others: NerfStudio, Albumentations, NetworkX, TensorFlow, OpenGL, Pandas, SciPy, Keras, OpenCV, Pillow/PIL, Transformers, Scikit-learn, PyTorch, NumPy, Matplotlib, NLTK, Tkinter, Scikit-image